

Deep Q Network (DQN), Rainbow DQN

$$Q^*(s, a) = r(s, a) + \gamma \mathbb{E}_{p(s'|s, a)} \max_{a'} Q^*(s', a')$$

$$\text{TD: } Q_{\text{new}}(s, a) = Q_{\text{old}}(s, a) + \underbrace{\underbrace{r(s, a) + \gamma \max_{a'} Q_{\text{old}}(s', a')}_{\text{target } y(s, a)} - Q_{\text{old}}(s, a)}_{\text{TD}}$$

$$F(Q) = \frac{1}{|S| \cdot |A|} \sum_{s,a} \frac{1}{2} (Q(s,a) - y(s,a))^2$$

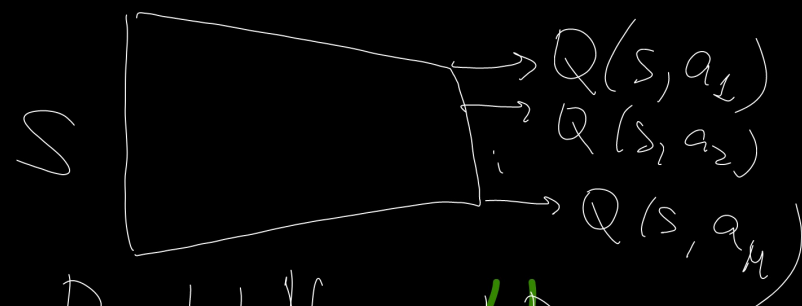
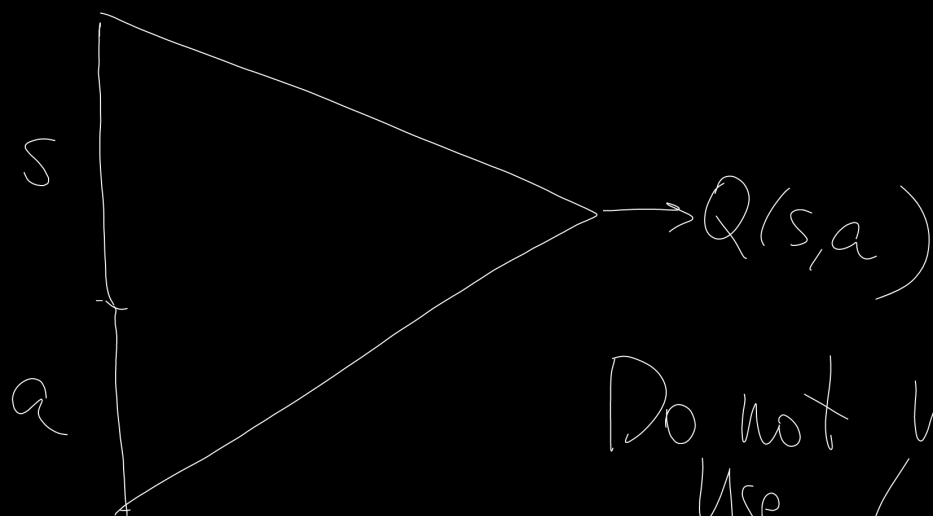
$$\sim s,a \quad Q_{\text{new}}(s,a) = Q_{\text{old}}(s,a) - \underset{\rightarrow \text{min}}{L} \frac{\partial F(Q)}{\partial Q} \underset{Q}{\quad}$$

$$= Q_{\text{old}}(s,a) - L (Q_{\text{old}}(s,a) - y(s,a))$$

= TD iteration

DQN: $Q(s, a | \theta) - NN$

$$F(\theta) = E_{s,a} \frac{1}{2} (Q(s, a | \theta) - y(s, a))^2 \rightarrow \min_{\theta}$$



Do not use Batch Norm / Drop Out
Use Layer Norm / Instance Norm instead

DQN scheme

Init. Θ , replay buffer M

for episodes:

Init. S_0

for $t = 0, 1, \dots, T$:

$a_t = \epsilon\text{-greedy}(Q(s_t; \Theta))$, get r_t, S_{t+1}

$(s_t, a_t, r_t, S_{t+1}) \xrightarrow{\text{store}} M$

$\{(s_i, a_i, r_i, S_{i+1})\}_{i=1}^B \xleftarrow{\text{sample}} M$

for each i : $y_i =$

$\begin{cases} r_i, & \text{if } S_{i+1} = \text{terminal} \\ r_i + \gamma \max_{a'} Q(s'_i, a' | \hat{\Theta}), & \text{otherwise} \end{cases}$

$\hat{\Theta}$ - target network

$$L(\theta) = \sum_i \frac{1}{2} (Q(s_i, a_i | \theta) - y_i)^2$$

$$\theta = \text{opt. step} (\nabla_{\theta} L)$$

$$\text{if } \text{iter} \% K == 0: \hat{\theta} = \theta$$

Rainbow DQN

1. DQN
2. n-step return
3. noisy networks
4. Double DQN
5. Dueling DQN
6. Prioritized replay
7. Distributional RL

n-step returns

$$n\text{-step return: } r_t + \gamma r_{t+1} + \dots + \gamma^{n-1} r_{t+n-1} +$$

$$\text{Store in buffer: } + \gamma^n \max_{a'} Q(s_{t+n}, a')$$

$$(s_t, a_t, r_t + \gamma r_{t+1} + \dots + \gamma^{n-1} r_{t+n-1}, s_{t+n})$$

Noisy Networks

$$z = Wx + b$$

Noisy network: $z = (M_w + S_w \odot \varepsilon_w)x + (m_b + s_b \odot \varepsilon_b)$

$$\text{Loss} = \mathbb{E}_{\varepsilon} \text{Loss}(M_w, S_w, m_b, s_b) + (m_b + s_b \odot \varepsilon_b)$$

$$\nabla \text{Loss} = \nabla \mathbb{E}_{\varepsilon} \dots = \mathbb{E}_{\varepsilon} \nabla \dots \rightarrow \min_{M, S, \dots}$$

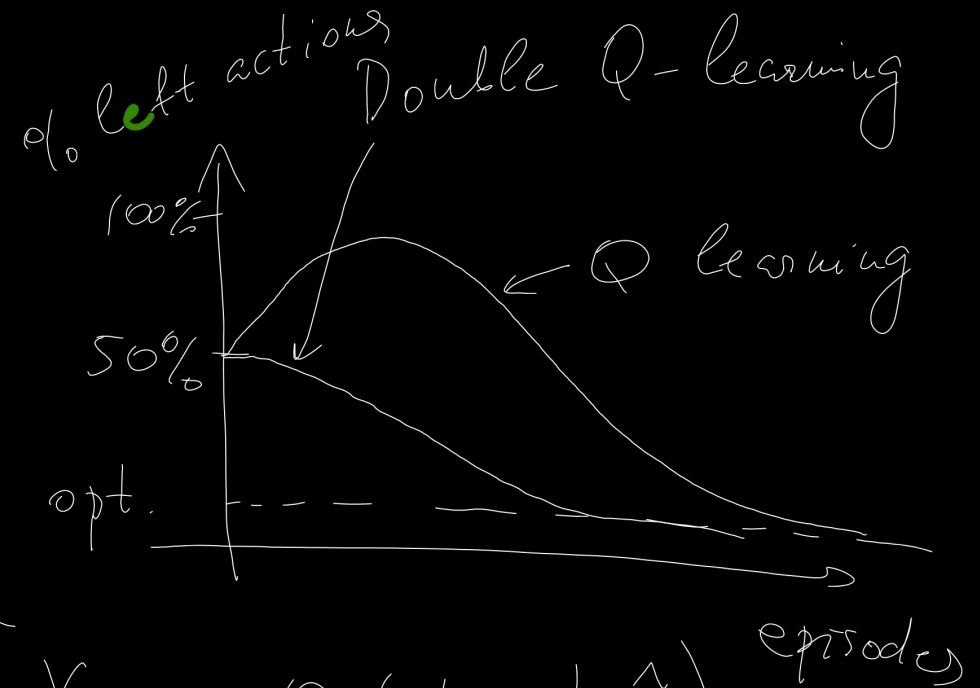
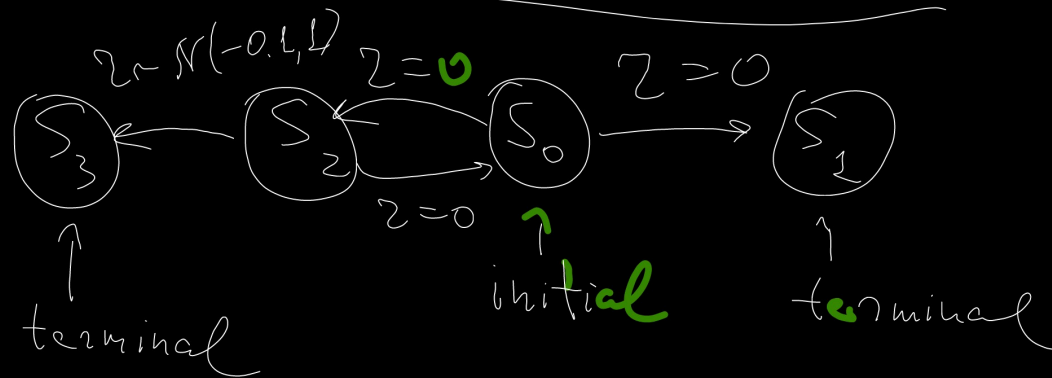
factored sampling

$$\varepsilon_{ij}^w = \varepsilon_i \cdot \tilde{\varepsilon}_j$$

$\varepsilon_i \sim \mathcal{N}(0, 1)$
for $i = 1, \dots, n_{row}$
 $\tilde{\varepsilon}_j \sim \mathcal{N}(0, 1)$
for $j = 1, \dots, n_{col}$

$$\varepsilon_w, \varepsilon_b \sim \text{Normal}$$

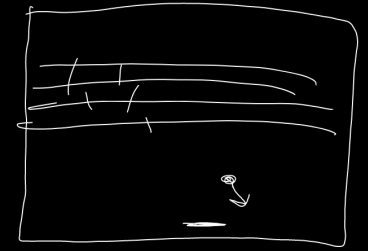
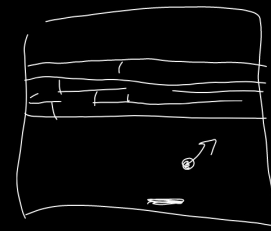
Double DQN



$$\text{DQN: target} = r + \gamma \max_{a'} Q(s', a' | \hat{\theta}) = r + \gamma Q(s', \arg \max_{a'} Q(s', a' | \hat{\theta}) | \hat{\theta})$$

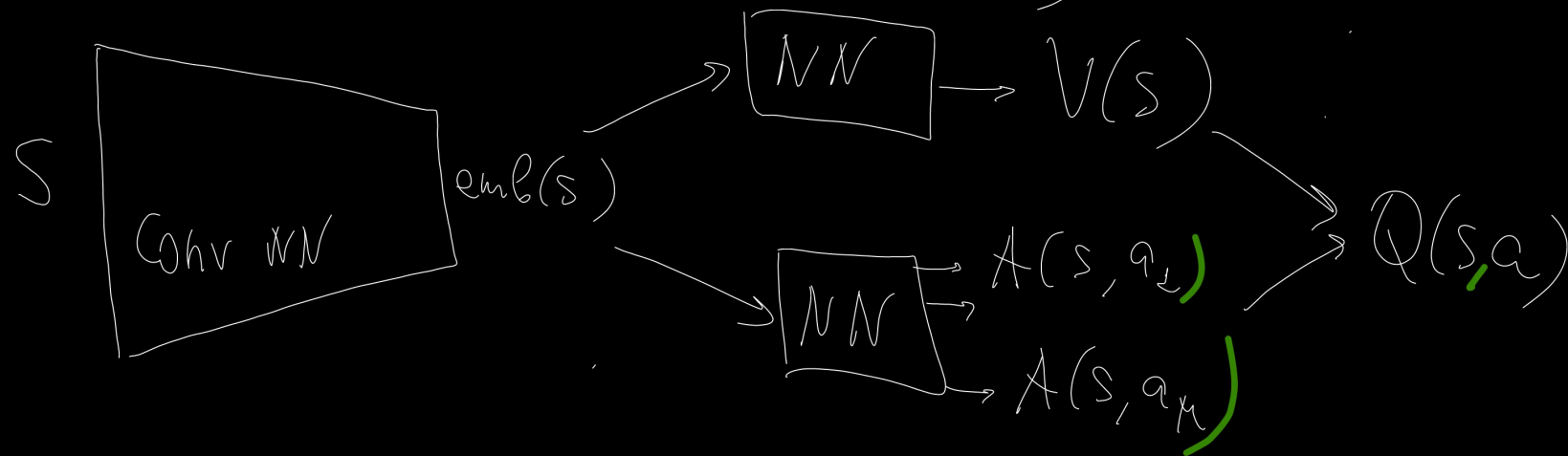
$$\text{Double DQN: target} = r + \gamma Q(s', \arg \max_{a'} Q(s', a' | \theta) | \hat{\theta})$$

Dueling DQN



$$A(s, a) \stackrel{\leftarrow \text{advantage}}{=} Q(s, a) - V(s) \Rightarrow$$

$$\Rightarrow Q(s, a) = A(s, a) + V(s)$$



$$Q(s, a) = V(s) + A(s, a) - \frac{1}{|A|} \sum_{a'} A(s, a')$$

$$V: 10 \quad A: -20$$

$$V: 20 \quad A: -30$$

$$V: -10 \quad A: 0 \leftarrow$$

Prioritized replay

$b = (s, a, r, s') \leftarrow \text{sample with prob.}$

$$p(b) \propto \underbrace{\left(|y(s, a) - Q(s, a)| + \epsilon \right)}_{TD}$$

SumTree data structure

$$O(\log N)$$

In loss add weights

$$\left(\frac{1}{N p(b)} \right)^B / \underset{\substack{\rightarrow \text{mini-batch}}}{\text{max } 15} \quad L \approx 0.5 \text{ in practice}$$

Distributional RL

$$Z^{\pi}(s, a) = r(s, a) + \gamma Z(s', a') + \dots$$

$$Q^{\pi}(s, a) = \mathbb{E} Z^{\pi}(s, a)$$

$$s' \sim p(s' | s, a)$$
$$a' \sim \pi(a' | s')$$

